MINI PROJECT REPORT

NEURAL NETWORK CLASSIFICATION PROJECT

Project Title: Iris Flower Classification Using Artificial Neural Network

1. Problem Statement

The aim of this project is to build an Artificial Neural Network (ANN) to classify Iris flowers into three species: Setosa, Versicolor, and Virginica. The model learns from flower measurements and predicts the correct flower species.

2. Dataset Description

The Iris dataset contains 150 flower samples and three flower species. Each sample has four features: sepal length, sepal width, petal length, and petal width. The dataset is used to train and test the neural network.

3. Data Preprocessing

The dataset was loaded using Python and scikit-learn. The input features and target labels were separated. The data was divided into 80% training data and 20% testing data. StandardScaler was used to bring the input features to a similar scale. This helps the neural network learn effectively.

4. Neural Network Architecture

The neural network was created using TensorFlow and Keras. It contains an input layer with four features, two hidden layers with 16 and 8 neurons, and an output layer with three neurons. ReLU activation is used in the hidden layers, while Softmax activation is used in the output layer to predict the flower species.

5. Model Training

The model was compiled using the Adam optimizer, sparse categorical cross-entropy loss, and accuracy as the evaluation metric. It was trained for 50 epochs. Training accuracy, validation accuracy, training loss, and validation loss were plotted to observe the learning process.

6. Results and Predictions

The trained model was tested using the test dataset. Test accuracy and test loss were calculated. The model generated predictions, and the predicted flower species were compared with the actual species.

Test Accuracy: _____ %

Test Loss: _____

A confusion matrix was created to display correct and incorrect predictions for all three flower species. The actual values were recorded after running the model.

7. Limitations

The dataset contains only 150 samples and three flower species. The model may not perform equally well on other flower datasets. Its results can also change depending on the training settings and data used.

8. Future Improvements

The model can be improved by testing different numbers of neurons and hidden layers, using more flower samples, and comparing different optimizers. Early stopping can also be used to reduce unnecessary training and help control overfitting.

9. Conclusion

This project demonstrates how to build a basic deep learning classification model using Python and TensorFlow/Keras. The process includes loading data, preprocessing, splitting the dataset, creating a neural network, training, prediction, evaluation, and visualization. Test accuracy, loss graphs, and the confusion matrix help us understand the model's performance on unseen data.